

calculated data concerning airflow and pressure drop

HOLTOP HBS-ZF1600/1600.500.10.0B

Engineering estimate for one heat recuperator in the worst expected summer operating condition

Status of this document: the pressure-drop values below are calculated / estimated design values. They are not a guaranteed manufacturer performance point for this exact unit. They are intended for fan sizing and system coordination until a supplier-certified curve is available.

1. Equipment basis

Parameter	Value / interpretation
Manufacturer	HOLTOP
Model	HBS-ZF1600/1600.500.10.0B
Nominal face dimensions	1600 x 1600 mm
Core depth	500 mm
Plate spacing	10 mm
Application considered	One recuperator in the process-air exhaust path
Worst summer airflow used for sizing	15,660 m³/h

2. Estimated pressure drop of one recuperator

For the selected HOLTOP exchanger, the following values are proposed for design purposes at the expected summer airflow.

Operating condition	Estimated pressure drop	Use
Normal, clean exchanger	100-150 Pa	Expected operating range
Worst summer condition, clean and dry	~150 Pa	Expected summer duty
Recommended engineering design value	200 Pa	Recommended value for fan selection calculations
Conservative allowance	250 Pa	Additional allowance for fouling and uncertainty

Recommended design assumption: use **200 Pa pressure drop** for one **HOLTOP recuperator** at the summer design airflow of approximately **15,660 m³/h**.

3. Reasoning behind the design value

The estimated range is based on the selected exchanger geometry, the relatively large 10 mm plate spacing and comparison with a similar cross-flow plate heat exchanger operating around the same order of airflow.

A previously reviewed comparable recuperator selection, with approximately 9.5 mm plate spacing and an airflow around 12,000 m³/h, showed pressure losses of approximately 130 Pa on the supply side and 133 Pa on the exhaust side. This is not the same exchanger and therefore is used only as a sanity check.

4. Summer operating condition

For the summer sizing case, the recuperator is expected to operate essentially dry. At an indicative process-air condition of about 44°C and 44% RH, the dew point is around 29°C. With summer outdoor air around 30°C, condensation inside the recuperator should therefore be limited or absent in the worst hot-weather case.

This is beneficial because condensation can partially block free passage area and increase pressure drop. Consequently, winter or cold-weather conditions may in practice produce a higher exchanger pressure loss than the hot, dry summer case, depending on humidity and surface temperatures.

5. Total pressure allowance for the final pull fan

For the final pull fan downstream of the recuperator, the recuperator itself is only one part of the total static pressure requirement.

System element	Design allowance
1 x HOLTOP recuperator	200 Pa
Transitions, bends and short duct sections	50-100 Pa
Roof discharge / outlet grille	20-50 Pa
Engineering reserve	50-100 Pa

Recommended fan selection point:
approximately **15,700 m³/h at 400 Pa static pressure.**

A practical selection range is approximately 350-450 Pa total external static pressure at 15,700 m³/h. The 400 Pa point provides a useful margin without intentionally oversizing the fan excessively.

6. Important system note

Only **one recuperator pressure drop** is included in the final pull-fan calculation. The two recuperators in the complete installation perform different functions and are not arranged in series in the same process-air stream.

7. Design value summary

Item	Design value
Summer design airflow	15,660 m³/h
Estimated clean recuperator pressure drop	~150 Pa
Recuperator design pressure drop	200 Pa
Conservative recuperator allowance	250 Pa
Recommended final pull-fan selection point	15,700 m³/h @ 400 Pa

8. Source / verification note

The geometry and product identification are based on the selected HOLTOP HBS-ZF1600/1600.500.10.0B configuration previously reviewed. The pressure-drop figures in this document remain engineering estimates until a manufacturer performance curve or certified pressure-drop value for this exact configuration is obtained.

